Essential in HVAC (40 Hours)

S.No	Session	Topics	Type	Duration (Hrs)
1	Session 1	**INTRODUCTION TO HEATING VENTILATION AND AIR CONDITIONING, PSYCHROMETRIC CHART:**	Theory/Demo	4.0 Hrs
		Types of Heat transfer modes		
		Air conditioning		
		Forms of Heat		
		BTU		
		Abbreviation's of standards		
		Important unit conversions		
		Psychrometric chart and temperature properties (RH, WBT, DBT, DP)		
		Psychrometric software		
2	Session 2	**REFRIGERATION CYCLE, TYPES OF AC SYSTEM:**	Theory/Demo	4.0 Hrs
		Vapour compression cycle		
		Basic refrigeration system		
		Refrigerants and their types		
		Components of refrigeration cycle		
		Types of AC		
		Classifications		
		Package AC		
		VRV system		
3	Session 3	**Cooling and Heating load calculation**	Theory/Demo	4.0 Hrs
		Indoor design conditions		
		Typical inside design conditions		
		Outside design data		
		Thermal resistance for materials		
4	Session 4	Overall heat transfer coefficient for components factor calculation	Theory/Demo	4.0 Hrs
		Heat load calculation		
		HAP software		
5	Session 5	**O general heat load calculation and Duct Designing:**	Theory/Demo	4.0 Hrs
		O general estimation		
		Ducts		
		Air Distribution system components		
		Types of pressure		
		Velocity in ducts		
		Equal friction method		
		Infiltration rate		
		Duct dimensioning		
		Duct Sizer		
6	Session 6	**AIR TERMINALS, VENTILATION TYPES**	Theory/Demo	4.0 Hrs
		Classification of ducts		
		Duct materials		
		Duct fittings		
		Plenum Designs duct design procedures		
		Fan Types		
		Types of ventilation		
		Exhaust air flow calculation		
		Static pressure calculation		
7	Session 7	**KITCHEN HOOD VENTILATION, CAR PARKING VENTILATION**	Theory/Demo	4.0 Hrs
		Design criteria		
		Wall type and island type designs		
		ESP		
		Kitchen Hood design properties and calculation		
		Types of car parking ventilation		
		Impulse and induction fans		
		Calculation of fresh air rate		
8	Session 8	**CHILLERS/DX SYSTEM**	Theory/Demo	4.0 Hrs
		Types of chillers		
		piping types		
		Valves		
		Pumps		
		Type of Heads		
		Chilled Water system		
		Pump selection criteria's		

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9	Session 9	**Cooling towers, Coil selection, VRF system, District cooling:**	Theory/Demo	4.0 Hrs
		Tower types and their characteristics		
		Example Calculations		
		Coils selection		
		Selection procedures		
		Variable refrigerant flow		
		Advantages of VRF		
		District cooling		
		Advantages		
		DC components		
		District cooling plant Equipment		
		Example calculation for room pressure		
10	Session 10	**DRAFTING USING AUTOCAD**	Theory/Demo	4.0 Hrs
		Kitchen Hood		
		Toilet ducting		
		AHU		
		Examples on car parking plans		
		Duct sizing in AutoCAD		
		Practise: Create smart features using knowledge tools. Practise workbook exercise and hands on exercise	Practicals	